Orthogonal Matrices, Adjoints and the Spectral Theorem

From the class book: Chapter 7, Sections 7.4–7.5

1. Key definitions

Adjoint. On a finite-dimensional inner product space every operator T has a unique adjoint T^* with $\langle Tu, v\rangle = \langle u, T^*v\rangle$. In an orthonormal basis $[T^*] = \overline{[T]}^{\,t}$. Rules: $(ST)^* = T^*S^*$, $(cT)^* = \bar{c}T^*$, $T^{**} = T$, $\ker T^* = (\operatorname{im} T)^{\perp}$.

Special operators. T is *self-adjoint* (Hermitian) if $T^* = T$, *skew-adjoint* if $T^* = -T$, *unitary* (orthogonal over $\mathbb{R}$) if $T^*T = I$, *normal* if $TT^* = T^*T$, and *positive* if $T^* = T$ and $\langle Tv, v\rangle > 0$ for $v \neq 0$.

- Hermitian, skew-Hermitian and unitary matrices are all normal; $\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$ is not normal.
- Every $A \in M_n(\mathbb{C})$ is $H + iK$ with $H = \frac{1}{2}(A + A^*)$, $K = \frac{1}{2i}(A - A^*)$ Hermitian; A is normal iff $HK = KH$.

2. Main results

Theorem (Spectral theorem)

- Complex: $A = UDU^*$ with U unitary and D diagonal $\iff$ A is normal.
- Real: $A = QDQ^t$ with Q orthogonal and D real diagonal $\iff$ A is symmetric.
- Spectral decomposition: $A = \sum \lambda_i P_i$, P_i the orthogonal projection onto E_{λ_i}, $\sum P_i = I$, $P_iP_j = 0$ $(i \neq j)$. Then $f(A) = \sum f(\lambda_i)P_i$.
- For normal T: $\|Tv\| = \|T^*v\|$, $\ker T = \ker T^*$, and eigenvectors for distinct eigenvalues are orthogonal.

Theorem (Isometries and singular values)

- Preserving inner products $\iff$ preserving norms $\iff$ preserving distances; on a finite-dimensional space this means unitary. A matrix is unitary iff its columns (or rows) are orthonormal.
- Real orthogonal: $\det = \pm 1$. A 3×3 orthogonal matrix with $\det = 1$ is a rotation: eigenvalues $1, e^{\pm i\theta}$, trace $1 + 2\cos\theta$.
- Singular values $\sigma_i = \sqrt{\lambda_i(A^*A)}$; $\max_{\|x\|=1} \|Ax\| = \sigma_1$ and $\sum \sigma_i^2 = \operatorname{tr}(A^*A) = \sum |a_{ij}|^2$.
- Schur: $\sum |\lambda_i|^2 \leq \sum |a_{ij}|^2$, with equality iff A is normal.

Normal matrix	Eigenvalues	Extra fact
Hermitian	real	normal with real eigenvalues $\Rightarrow$ Hermitian
skew-Hermitian	in $i\mathbb{R}$	e^A is unitary
unitary	$\lvert\lambda\rvert = 1$	Hermitian and unitary $\Rightarrow A^2 = I$
positive definite	> 0	unique positive square root
normal nilpotent	0	it is 0

3. Methods

Method: orthogonal diagonalization of a real symmetric A.

- (1) Find the eigenvalues; split into blocks when A is block diagonal.
- (2) For each eigenvalue, find a basis of E_λ and apply Gram–Schmidt inside it (different eigenspaces are already orthogonal).
- (3) Put the unit vectors as the columns of Q: $Q^tAQ = D$.

Method: functions of a normal matrix.

- Write $A = \sum \lambda_i P_i$, with $P_i = \frac{v_i v_i^*}{v_i^* v_i}$ for a simple eigenvalue.
- Then $A^n = \sum \lambda_i^n P_i$, $\sqrt{A} = \sum \sqrt{\lambda_i} P_i$, $e^A = \sum e^{\lambda_i} P_i$.

4. Exam shortcuts and common traps

- To test normality fast, compare $\sum \lvert a_{ij}\rvert^2$ with $\sum \lvert\lambda_i\rvert^2$.
- Unitary (or normal) and triangular $\Rightarrow$ diagonal. A real skew-symmetric orthogonal matrix has eigenvalues $\pm i$, so its order is even.
- If A is Hermitian then $A \pm iI$ is invertible, and $(A - iI)(A + iI)^{-1}$ is unitary (Cayley transform).
- **Traps:** a complex *symmetric* matrix need not be Hermitian; a real orthogonal matrix such as a rotation is unitarily diagonalizable over $\mathbb{C}$ but not orthogonally over $\mathbb{R}$; a matrix with orthogonal (not orthonormal) columns is not orthogonal; singular values are not $\lvert\lambda_i\rvert$ unless A is normal.

5. Solved examples

Example 1. Orthogonally diagonalize $A = \begin{pmatrix} 3 & 0 & 0 \\ 0 & 6 & 2 \\ 0 & 2 & 6 \end{pmatrix}$.

Solution

- The block $\begin{pmatrix} 6 & 2 \\ 2 & 6 \end{pmatrix}$ has eigenvalues $8, 4$ with eigenvectors $(1,1)$, $(1,-1)$.
- With columns $\frac{1}{\sqrt{2}}(0,1,1)$, $\frac{1}{\sqrt{2}}(0,1,-1)$, $(1,0,0)$: $Q^t A Q = \operatorname{diag}(8,4,3)$.

Example 2. Is $A = \begin{pmatrix} 1 & i \\ i & 1 \end{pmatrix}$ Hermitian? Normal? Find its eigenvalues.

Solution

- $A^* = \begin{pmatrix} 1 & -i \\ -i & 1 \end{pmatrix} \neq A$: not Hermitian, although $A^t = A$.
- $AA^* = A^*A = 2I$: normal (indeed $A/\sqrt{2}$ is unitary). Eigenvalues $1 \pm i$, not real.

Example 3. For $A = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$, find the spectral decomposition, A^n and the positive square root.

Solution

- $\lambda = 3, 1$; $P_1 = \frac{1}{2}\begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$, $P_2 = \frac{1}{2}\begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix}$, $A = 3P_1 + P_2$.
- $A^n = 3^n P_1 + P_2$ and $\sqrt{A} = \sqrt{3}\,P_1 + P_2$.

Example 4. Find the singular values of $A = \begin{pmatrix} 3 & 0 \\ 4 & 5 \end{pmatrix}$ and $\max_{\|x\|=1} \|Ax\|$.

Solution

- $A^tA = \begin{pmatrix} 25 & 20 \\ 20 & 25 \end{pmatrix}$ has eigenvalues $45, 5$: $\sigma = 3\sqrt{5},\ \sqrt{5}$.
- The maximum is $3\sqrt{5}$. Check: $45 + 5 = 50 = 9 + 16 + 25$. The eigenvalues $3, 5$ differ, since A is not normal.

Example 5. A 3×3 real orthogonal A has $\det A = 1$ and $\operatorname{tr} A = 2$. Find its eigenvalues.

Solution

- A is a rotation: $1 + 2\cos\theta = 2$, so $\theta = \pi/3$.
- The eigenvalues are $1,\ e^{\pm i\pi/3} = \frac{1}{2} \pm \frac{\sqrt{3}}{2}i$.

Example 6. Let $A \in M_n(\mathbb{C})$ be normal. Show that (a) real eigenvalues force $A^* = A$, and (b) $A^k = 0$ forces $A = 0$.

Solution

- $A = UDU^*$. (a) D real gives $A^* = UD^*U^* = UDU^* = A$.
- (b) $D^k = 0$ gives $D = 0$, so $A = 0$. Hence $\begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$ is not normal.